



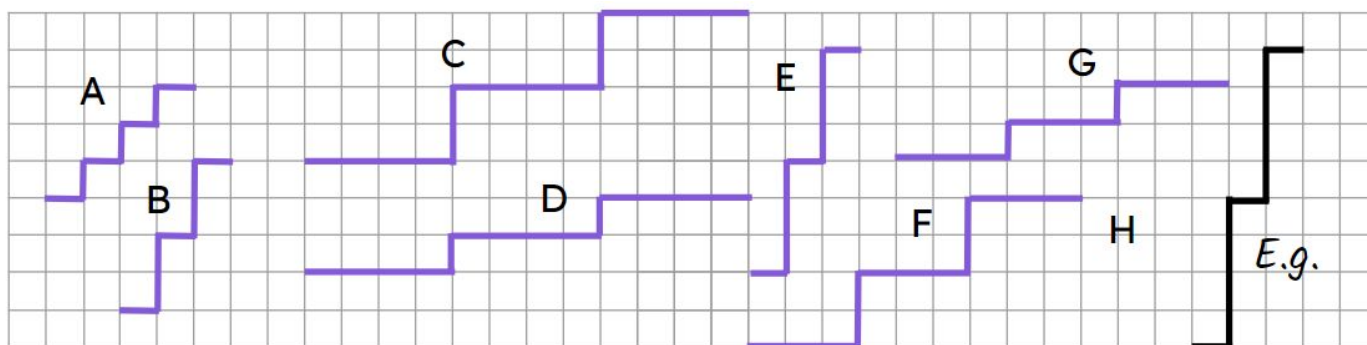
Positive rate of change from a graph

Task A: Defining gradient

1)

a) Put these steps in order from steepest to gentlest.

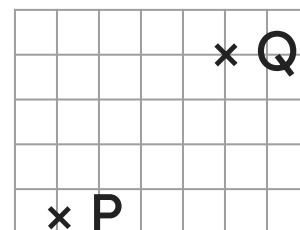
steepest: E - B - A - F - C - G - D gentlest



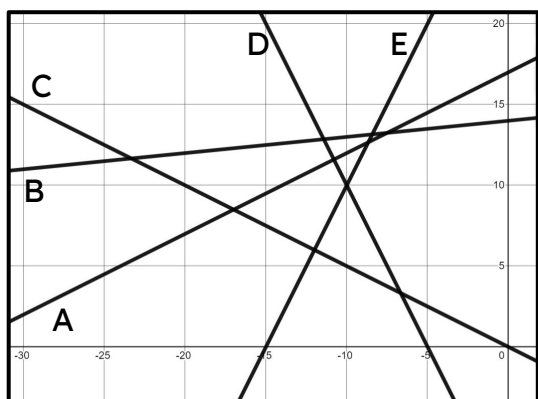
b) Draw staircase H so it is the steepest staircase.

c) Another set of steps join point P to point Q. Where would these steps go in order of steepness?

Same as A



2) Which lines have a positive **gradient**?



A, B and E have positive gradients

3) Which tables of values show a relationship which will have positive **gradient** when plotted?

a)

x	0	1	2	3
y	3	5	7	9

Positive, y increases as x increases

b)

x	-2	-1	0	1
y	12	8	4	0

Not positive, y decreases as x increases

c)

x	-8	-7	-6	-5
y	-8	-5	-2	1

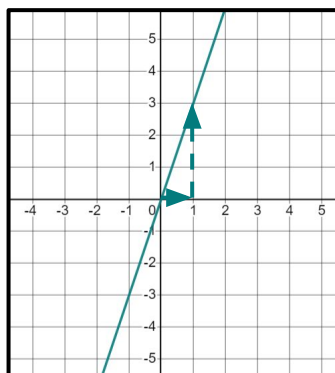
Positive, y increases as x increases



Task B: Calculating positive gradients from graphs

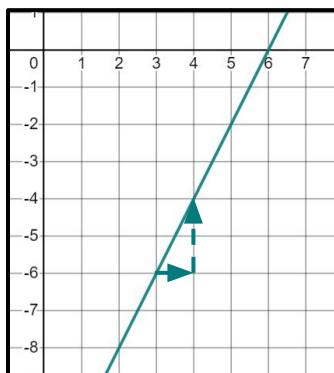
1) Find the **gradient** of each line. You can draw a ratio table to help you.

a)



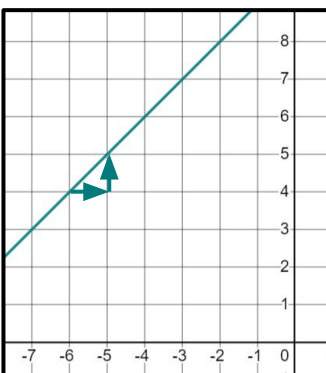
3

b)



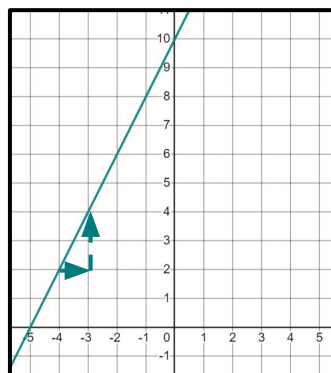
2

c)



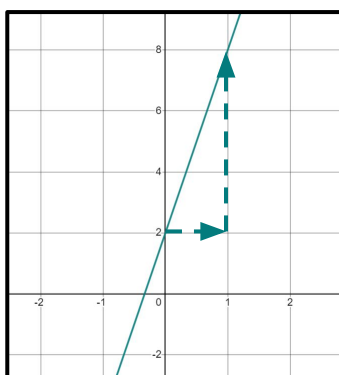
1

d)



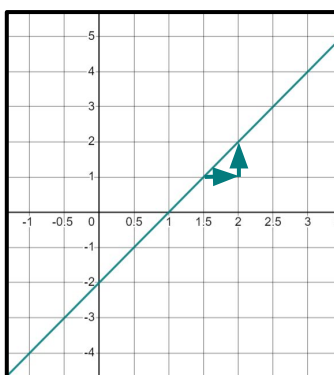
2

e)



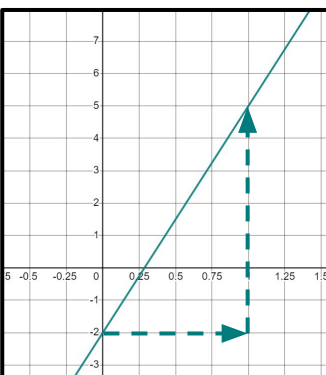
6

f)



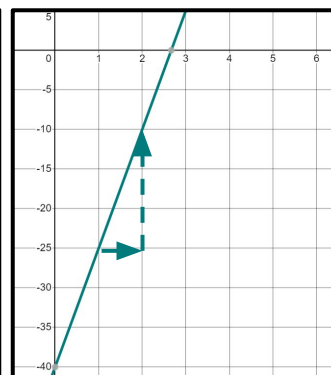
2

g)



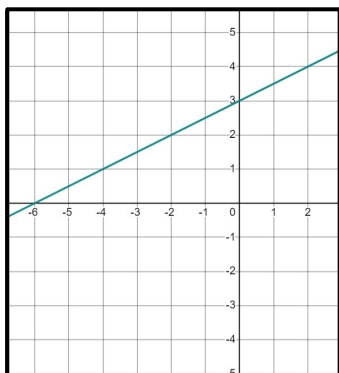
7

h)

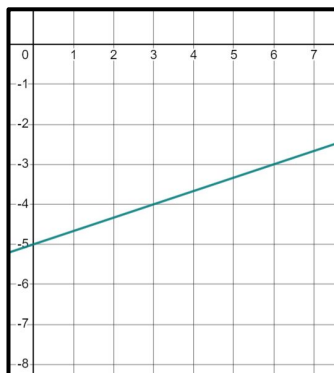


15

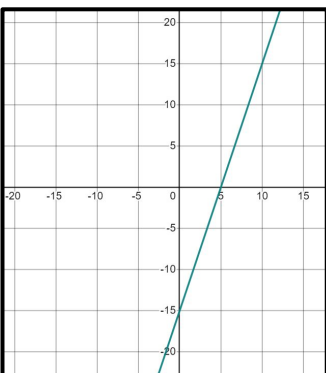
i)



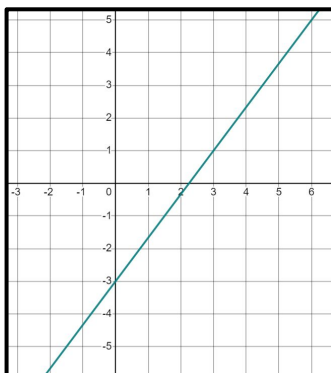
j)



k)



l)



2	1
1	$\frac{1}{2}$

3	1
1	$\frac{1}{3}$

5	15
1	3

3	4
1	$\frac{4}{3}$